CS-670 Course Project Report

Privacy, Representation, and Noise Propagation in Federated Recommendation

A Reproducible Study of Low-Rank Models, Local Personalization, and User-Level Differential Privacy

Bhavya Chauhan (230294)
Chaudhari Divyesh (230325)
Avantika Rohite (230245)
Aaradhya Rohi (230011)

Final project report

Abstract

Federated recommendation separates shared item learning from local user personalization, but whole-user differential privacy (DP) requires clipping and noisy aggregation that can degrade recommendations. This project studies whether a smaller shared representation improves that trade-off and how injected noise becomes disturbance to item matrices, personalized scores, and rankings. A reproducible MovieLens-100K pipeline compares centralized Bayesian Personalized Ranking (BPR), a user-weighted control, and federated full-matrix and low-rank models with and without DP. Low-rank private models beat the private full-matrix reference in four of sixteen multiplicity-adjusted comparisons, while all four nonprivate low-rank variants fall below the converged full model.

A separately frozen, equal-budget extension finds that fixing a public factor obtains no positive adjusted comparison and loses eight of sixteen to joint factor training; a user-level private popularity baseline exceeds every collaborative point estimate at target $\epsilon \leq 4$. Conditional moments and measured perturbations show that effective noise depends on factor geometry, local user scale, clipping, and pairwise margins, not on coordinate norms alone. Pulse-and-reset studies reveal large two-factor trajectory differences that an orthogonal alignment control substantially reduces; thirty transferred MovieLens-1M training runs and 200 further probes reproduce this dependence on how noise is coupled across rebalanced factors. A screen of 3,200 item-pair records finds a maximum absolute sign-flip probability error of 7.67×10^{-6} for a variance-matched Gaussian approximation, rejecting non-Gaussian tails as a practically consequential explanation in the sampled states.

Keywords: federated recommendation; user-level differential privacy; matrix factorization; BPR; effective noise; personalization; factor alignment.

Contents

Abstract i
1 Introduction 1
2 Related work 2
3 Problem formulation and privacy model 3
4 Models 4
5 Evaluation protocol and experimental design 6
6 Results of the baselines 8
7 E1: effective noise and fixed-factor training 10
8 E2/E2b: local-state memory and noise coupling 12
9 E3: larger-data replication and balancing controls 13
10 E4: a conditional ranking-tail screen 15
11 Discussion and conclusions 17
References 19
A Mathematical derivations 21
B Algorithm and settings 22
C Disclosures affecting interpretation 23
D Provenance and reproduction 24

1 Introduction

Recommendation systems use interaction histories that can reveal preferences users may not want to disclose. Federated learning keeps those records on clients while a server coordinates a shared model, but transmitted updates can still depend strongly on an individual's records. Differential privacy constrains that dependence. Here the protected unit is an entire user, so protecting one rating while exposing the rest of a history would be insufficient.

Privacy also changes the learning problem. Each client update is clipped, and random noise is added to the aggregate. Clipping can suppress informative directions; noise can obscure weak collaborative signals. These effects interact with client sampling, local training, item representation, and persistent user embeddings, so not every difference from a centralized model is a privacy cost.

The starting hypothesis was that fewer shared item parameters would improve the privacy–utility–communication trade-off. A full item matrix $Q \in \mathbb{R}^{M\times d}$ can be approximated by factors $A \in \mathbb{R}^{M\times r}$ and $B \in \mathbb{R}^{r\times d}$ with $r < d$, reducing communicated scalars from Md to $r(M + d)$. Under isotropic Gaussian perturbation, fewer coordinates also reduce the expected squared noise norm in those coordinates. Neither fact implies that scores are less disturbed: the model uses the product AB, and predictions multiply it by a local user vector whose scale can change during noisy training.

Research questions.

RQ1. **Representation and utility.** At matched privacy targets and a fixed private horizon, when does a low-rank shared representation outperform a full matrix, and what utility is lost before privacy is introduced?

RQ2. **Parameterization and effective noise.** Does fixing one public factor improve prediction quality once initialization, tuning opportunity, clipping controls, and a private popularity reference are included?

RQ3. **Persistent personalization.** After a shared-model perturbation has influenced local user states, how much prediction disturbance remains once the shared state is restored?

RQ4. **Validity of trajectory comparisons.** How much of a paired two-factor trajectory difference depends on rebalancing and on the rule coupling noise between equivalent factor representations?

RQ5. **Ranking-tail relevance.** In trained states, does the exact conditional law of a noisy item-pair margin differ enough from Gaussian approximations to motivate a different ranking method?

These questions concern distinct quantities: a difference in score RMS does not imply a difference in relevance, a change in the top-10 set need not lower NDCG, and a smaller transmitted tensor does not establish lower communication to convergence.

Contributions. (i) A reproducible experimental framework: deterministic data preparation, full-catalog ranking, explicit user-level aggregation and accounting, matched controls, and repeated seeds. (ii) A controlled evaluation of effective noise relating factor-space noise to the item product, local user state, clipping, ranking, and communication, benchmarked against a separately calibrated private popularity method. (iii) A diagnostic finding, replicated on a second MovieLens dataset, that large paired distances in rebalanced two-factor trajectories can

depend strongly on coordinate coupling. (iv) A tested and rejected mechanistic candidate: a non-Gaussian margin law whose approximation error is negligible in the selected trained states.

Stages and labels. The baselines B0–B4 step from centralized BPR (B0, with user-weighted control B0-UW) to the federated full matrix without and with DP (B1, B2) and jointly trained low-rank factors without and with DP (B3, B4). E1 then compares three separately named parameterizations under equal tuning: Full, Two (both factors trained), and FixedB (one public factor fixed). E2/E2b probe persistence and coupling, E3 replicates on MovieLens-1M, and E4 screens ranking tails. Sections 4 and 5 give the details.

2 Related work

Ranking, federation, and privacy. BPR learns implicit-feedback rankings by favoring observed over sampled unobserved items [1]; with matrix-factorization scores its shared and local components are easy to identify and audit, and it serves here as a diagnostic baseline rather than the strongest contemporary recommender. Federated averaging motivates repeated local optimization and update aggregation [2]; here clients are users with persistent vectors, and training is simulated centrally. DP-SGD and user-level federated mechanisms supply the bounded-contribution Gaussian construction [3, 4]. Accounting uses numerical privacy-random-variable (PRV) composition, with Rényi DP (RDP) as a cross-check [5–7]; the popularity baseline uses analytic Gaussian calibration [8]. Clipping has its own effects on convergence that noise norms cannot reveal [9].

Private recommendation and personalization. Private alternating least squares (ALS) perturbs sufficient statistics for matrix completion [10]; federated matrix factorization [11] and PrivateRec [12] address related settings with different objectives, privacy units, and trust assumptions. These were not implemented, so this report makes no comparison against them; they define the next comparators (section 11). Local/global personalization has a theoretical foundation under user-level joint DP [13], and public item features can reduce reliance on private observations [14]. FixedB in this project differs from the latter: its basis is random and contains no semantic movie information.

Low-rank geometry and alignment. Close precedents for E1's fixed-factor design are FFA-LoRA [15], which studies a fixed-factor strategy in private federated low-rank adaptation, and FedASK [16], which addresses limitations of fixed-matrix approaches; their applications and privacy settings differ from whole-user BPR recommendation. The geometry of DP perturbations has wider precedents [17]: PRISM studies gauge-dependent low-rank noise [18], FLoRG uses Procrustes alignment against decomposition drift [19], and FedRot-LoRA and FedGSA address rotational alignment and geometry-consistent aggregation [20, 21]. E2b applies this established alignment idea to test how a paired-noise construction affects personalized recommendation diagnostics.

Noise propagation and ranking stability. Earlier noise influences later gradients: NoiseCurve studies cross-iteration noise through model curvature [22], and DiSK develops noise reduction for private optimization [23]. The question here is narrower, concerning persistent local user state and paired factorized trajectories. Ranking stability under noisy scores is also active

[24], and products of Gaussians have established distributional theory [25], which E4 applies to a conditional BPR margin. The literature review was a targeted primary-source audit rather than an exhaustive systematic review.

3 Problem formulation and privacy model

3.1 Personalized implicit-feedback ranking

Let $\mathcal{I}_u^+$ be user u's training-positive items. A local vector $p_u \in \mathbb{R}^d$ and shared item row $q_i \in \mathbb{R}^d$ produce the score $s_{ui} = p_u^T q_i$, so that $S = PQ^T$ (no item biases, content, or temporal features). For a triplet (u, i, j) with $i \in \mathcal{I}_u^+$ and $j \notin \mathcal{I}_u^+$, the BPR loss is

$$\ell(u, i, j) = \log(1 + \exp(s_{uj} - s_{ui})) + \lambda(\|p_u\|^2 + \|q_i\|^2 + \|q_j\|^2). \tag{1}$$

In a factorized model $q_i = A_i B$, and gradients flow through this effective row. Interaction-weighted training gives more weight to users with more positives; user-weighted training averages within each user first. Uniform client inclusion is closer to the latter, which is why the B0-UW control exists.

3.2 Protected information and adjacency

The privacy unit is one user's complete record under add/remove adjacency. A mechanism $\mathcal{M}$ is (ϵ, δ)-DP when, for neighboring D, D' and each measurable set $\mathcal{O}$,

$$\Pr[\mathcal{M}(D) \in \mathcal{O}] \leq e^{\epsilon} \Pr[\mathcal{M}(D') \in \mathcal{O}] + \delta. \tag{2}$$

The catalog, population size used for normalization, configurations, and random public basis are fixed public inputs; eligibility filtering and catalog construction are not privatized. User vectors stay local and are never in the shared update. The trust model assumes a server that sees only a noised aggregate, as secure aggregation would support; the simulator does not implement that protocol.

3.3 Private aggregation mechanism

Let θ_t be the shared parameters: vectorized Q, (A, B), or A alone. Each round, clients are sampled independently with probability q; a selected client updates locally from θ_t and returns the shared difference $\Delta_{u,t}$. The whole contribution is clipped once:

$$\bar{\Delta}_{u,t} = \Delta_{u,t} \min\left(1, \frac{C}{\|\Delta_{u,t}\|_2}\right), \ \theta_{t+1} = \theta_t + \frac{\eta_s}{qN}\left(\sum_{u \in \mathcal{U}_t} \bar{\Delta}_{u,t} + \xi_t\right), \ \xi_t \sim \mathcal{N}(0, \sigma^2 C^2 I). \tag{3}$$

For Two, the norm covers the concatenated factor update. Noise reaches every shared coordinate, the denominator is the fixed qN, and empty rounds still release noise and count toward composition (appendix B.1 gives the exact client procedure). At the ML-100K population $qN = 94.2$; at ML-1M, $qN = 603.5$. The accountant depends only on q, T, σ, δ, so transferring these parameters preserves the bound while changing the post-normalization noise SD. Changing dimension alone does not reduce ϵ, although utility can change substantially.

All final private collaborative experiments use $T = 50$, $q = 0.1$, $C = 1$, and $\delta = 10^{-5}$ (table 1). PRV is the declared primary bound.

Table 1: Final iterative privacy calibration at $q = 0.1$, $T = 50$, $\delta = 10^{-5}$.

ϵ target	σ	PRV bound	RDP cross-check
8	0.805072021	7.967513	9.119818
4	1.151928711	3.963885	4.497283
2	1.760449219	1.993278	2.220645
1	2.977490234	0.990062	1.087196

PRV is primary. RDP is a separate bound on the same mechanism. Source: privacy_accounting.csv, final 50-round rows.

Release scope. The guarantee covers one fixed-hyperparameter training mechanism. Publishing several runs requires composition, and private validation-based selection and user-level diagnostics are outside it. Local personalization fits the joint-DP framework when a user's output depends on their own records and an appropriately private shared transcript; this does not authorize publishing the matrix P. E2 and E3 use oracle reset and alignment operations on private states and E4 conditions on private checkpoints: these are research measurements, not releases covered by the checkpoint's privacy label. Reproducible seeds support auditing but are not production-grade secure randomness.

3.4 Notation and conventions

N, M	eligible users; catalog items	q, E, T	inclusion probability; local epochs; rounds
d, r	embedding dim. (64); rank (4,8,16,32)	C, σ	clipping threshold; noise multiplier
P, Q	user matrix; effective item matrix	η_ℓ, η_s	local and server learning rates
A, B	item factors, $Q = AB$	τ	coordinate noise SD, $\eta_s \sigma C/(qN)$

A value reported as mean $\pm$ SD describes variation across seeds, not a confidence interval. Smaller ϵ means stronger privacy; labels $\epsilon = 1, 2, 4, 8$ denote targets, with accountant bounds in table 1. Top-10 churn is one minus top-10 set overlap divided by ten, averaged over observer users.

4 Models

All final settings are listed in appendix B.2.

B0 and B0-UW: centralized controls. B0 learns 64-dimensional user and item embeddings centrally with BPR, sampling interactions uniformly. B0-UW keeps the same model and optimizer but samples a user uniformly and then a positive within that user. Together they separate the change in user weighting from the change to federated optimization.

B1: federated full matrix without DP. Users are included with $q = 0.1$; each selected user starts from the shared Q and its persistent vector and runs two local epochs. Item differences are averaged with equal client weight over realized participants. B1 restores its best validation checkpoint and is the converged nonprivate reference.

B2: private full matrix. B2 applies equation (3) to the whole item update for 50 rounds, evaluating the final round without privacy-dependent early stopping. A matched 50-round no-DP control is needed because the gap from converged B1 includes a training-budget effect even without noise.

B3 and B4: jointly trained factors. B3 represents $Q = AB$ at ranks $4, 8, 16, 32$ without DP, regularizing p_u and the effective rows $A_i B$. QR decompositions and a core SVD rebalance the factors while preserving their product up to floating-point tolerance. This prevents extreme reciprocal scaling but cannot remove a perturbation already in AB (appendix A.3), and it introduces a changing orthogonal coordinate convention that becomes central in E2 and E3. B4 applies one joint clip to $(\Delta A, \Delta B)$ and independent Gaussian coordinates across the concatenated vector, with the same q, T, C, σ, δ as B2. B4 received more validation tuning than B2 (appendix C), so this compares retained methods rather than intervening on coordinate count alone.

E1: Full, Two, and FixedB. E1 is a separately frozen comparison. Full repeats the final full-matrix mechanism; Two trains both factors; FixedB holds a public row-orthonormal B, seeded independently of interactions, and privately trains only A. At each rank Two and FixedB begin from the same effective Q. Each of nine units (Full and two methods at four ranks) receives the same four local/server rate candidates: $(1.25, 1)$, $(2.5, 1)$, $(5, 0.5)$, and $(5, 1)$. Selection uses three seeds at reference target four, subject to finite behavior across six levels: no clipping/no noise, clipping only, and the four private targets. FixedB's selected local rate reaches the upper grid boundary, which limits claims about its optimal attainable utility. FixedB keeps the same hard score-rank bound as Two at rank r (appendix A.4), so its results cannot be attributed simply to a smaller maximum score rank.

User-level private popularity. For each user take a binary indicator $h_u \in \{0, 1\}^M$ of distinct training-positive items and release

$$x_u = h_u \min\left(1, \frac{\sqrt{20}}{\|h_u\|_2}\right), \qquad z = \sum_u x_u + \mathcal{N}(0, 20\sigma_{\text{pop}}^2 I). \tag{4}$$

One full-population release is calibrated analytically at the same (ϵ, δ) (appendix A.7), and items are ranked by z_i under the same candidate filter. Raw popularity, bounded no-noise popularity, and private popularity are distinct references, since the norm bound downweights high-activity users.

Communication accounting. The simulator counts dense float32 uploads and downloads for participating clients, excluding local P. The payload per client-round is

$$b_{\text{Full}} = 8Md, \qquad b_{\text{Two}} = 8r(M + d), \qquad b_{\text{FixedB}} = 8Mr \quad \text{bytes}. \tag{5}$$

Network framing, compression, cryptographic overhead, and device latency are not measured.

Table 2: Dataset sizes under the frozen chronological implicit-feedback protocol.

Quantity	MovieLens-100K	MovieLens-1M
Eligible users	942	6,035
Catalog items appearing in ratings	1,682	3,706
Training positives	53,491	563,206
Validation / test targets	942 / 942	6,035 / 6,035
Test targets scored in this project	942	0

The ML-100K raw release contains 943 users and 100,000 ratings; 55,375 ratings are positive. ML-1M has 575,281 positives before eligible-user filtering; five users are removed. The ML-1M catalog is the set represented in the ratings file.

5 Evaluation protocol and experimental design

5.1 Datasets and splitting

The benchmark is the official MovieLens-100K release [26]; E3 applies the unchanged pipeline to MovieLens-1M (table 2). Ratings of four or five are positive implicit feedback; lower ratings are not training positives but still filter candidates. Users need at least three positives. Each user's positives are ordered by timestamp: the last is the test target, the penultimate the validation target, and the rest training, with equal timestamps broken by a deterministic SHA-256 rule. This is a per-user chronology, not a global temporal split. Ties cannot recover the true order of simultaneous events: on ML-100K, 316 users have a validation timestamp equal to their final training timestamp, and 305 have equal validation and test timestamps, so this convention is consequential.

Activity groups come from ML-100K training counts: low has at most 22 positives (319 users), medium 23–61 (310), and high at least 62 (313). Six validation and fourteen test targets are cold items with no training positive in the population. They remain in the evaluation, but they are too few for a reliable cold-item claim.

5.2 Sampling, candidates, and metrics

Training negatives are uniform over items outside the user's training positives and never consult held-out labels, so a future positive or a low-rated item can occasionally be sampled; filtering it out would use held-out information. For each held-out event, evaluation ranks the complete catalog after removing everything the user rated before that event, keeping the target; there is no sampled-negative evaluation. Later ratings never filter an earlier event, so validation and test use different candidate filters. Metric ties are resolved pessimistically. Diagnostic top-10 sets in E2–E4 instead exclude training positives only and are label-free. With one held-out item of rank R_u,

$$\mathrm{NDCG@10}_u = \frac{\mathbb{1}\{R_u \le 10\}}{\log_2(R_u + 1)}, \quad \mathrm{HR@10}_u = \mathbb{1}\{R_u \le 10\}, \quad \mathrm{MRR@10}_u = \frac{\mathbb{1}\{R_u \le 10\}}{R_u}, \tag{6}$$

averaged equally over users. NDCG@10 is the primary measure; HR@10 and MRR@10 are secondary, without confirmatory claims, and Recall@10 equals HR@10 in this one-target design.

Holdout exposure. ML-100K test results were used across stages, so it is a repeatedly used development benchmark rather than a pristine holdout. Later stages freeze their choices before

Table 3: Experimental stages and their evidence units.

Stage	Question	Retained evidence
Phase 0	Can the ranking protocol be reproduced?	Data fingerprints, evaluator checks, random and raw-popularity references.
B0–B4	How do weighting, federation, privacy, and rank interact?	Three-seed weighting control; five-seed final federated comparisons.
E1	Does a fixed public factor help under equal tuning?	648 search jobs; 270 final collaborative states; 21 popularity arrays.
E2/E2b	Is local-state disturbance persistent and coupling-sensitive?	120 pulse/reset probes and 40 aligned follow-up probes from 30 checkpoints.
E3	Does the diagnostic replicate and depend on balancing?	30 ML-1M training runs; 200 controlled probes across two datasets.
E4	Do non-Gaussian margins matter in trained states?	50 states; 3,200 prescribed item pairs; six synthetic cases.

A probe is not an independently trained model. E2–E4 deliberately reuse frozen checkpoints.

scoring. For ML-1M, validation has been evaluated for the 30 trained models and test targets remain unscored.

5.3 Experimental stages

The stages in table 3 build on each other, with explicit controls for competing explanations: B0-UW addresses weighting, 50-round no-noise controls address undertraining, clipping-only controls separate clipping from perturbation, and private popularity tests whether iterative personalization is competitive at all. E1's grid of $9 \times 4 \times 3 \times 6 = 648$ search jobs includes eleven retained failures, and its selections were frozen before the first extension test scoring.

5.4 Statistical analysis

The weighting comparison uses seeds 42, 123, and 2026; final B1–B4 and E1 add 7 and 99. To compare methods on NDCG@10, the metric is averaged over seeds for each user, and the paired user differences are bootstrapped 100,000 times. The primary families are four B3-versus-B1 comparisons (98.75% intervals), sixteen B4-versus-B2 comparisons, and sixteen matched-rank FixedB-minus-Two comparisons (99.6875% each), all Bonferroni-adjusted. These intervals condition on the saved training runs. An interval above zero supports a positive comparison; one containing zero is recorded as such, not as equivalence. Ordinary 95% intervals for secondary comparisons do not carry this multiplicity interpretation. Figure whiskers are seed SDs and are not the decision criterion.

Low-rank curves select each method's rank on validation at each noise level; these differ from the matched-rank primary comparisons, and a test-best rank never replaces a validation-selected one. Activity groups and personalization checks are descriptive. E2–E3 average two pulse draws within a seed before summarizing across five seeds, also descriptively. E4 declares its checkpoints, pair types, and practical-error gate before examining outputs. Provenance, code entry points, and rebuild steps are in appendix D.

Table 4: Converged nonprivate federated utility (five seeds).

Model	NDCG@10 ± seed SD	Adjusted interval vs. B1
B1	0.08901 ± 0.00249	Reference
B3 r4	0.05947 ± 0.00381	[-0.041779, -0.017893]
B3 r8	0.07101 ± 0.00596	[-0.027511, -0.008949]
B3 r16	0.07633 ± 0.00722	[-0.020950, -0.004560]
B3 r32	0.06807 ± 0.01354	[-0.031093, -0.011160]

Intervals are 98.75% paired-user bootstrap intervals for the four-comparison B3 family. Source: final_5seed_means.csv and paired_contrasts.csv.

Table 5: Historical private and matched no-DP test NDCG@10 (five-seed means).

Target / control	B2 Full	B4 r4	B4 r8	B4 r16	B4 r32
No DP	0.057889	0.040834	0.048083	0.049143	0.050931
8	0.052271	0.038544	0.045385	0.043628	0.044022
4	0.042169	0.037846	0.042653	0.038274	0.044489
2	0.025428	0.029079	0.034485	0.033993	0.036504
1	0.011461	0.015976	0.018162	0.016685	0.015390

All rows use 50 rounds. Historical tuning opportunities differ. Seed SDs and secondary metrics appear in the supplement. Source: final_5seed_means.csv.

6 Results of the baselines

6.1 Sanity references and centralized learning

Under the common full-catalog evaluator, random ranking reaches approximately 0.0038 test NDCG@10 and raw popularity 0.044292. In the matched three-seed comparison, B0 reaches 0.085972, B0-UW 0.091745, and B1 0.087539. The paired differences B0-UW$-$B0 (0.005773, 95% interval $[-0.000164, 0.011910]$), B1$-$B0-UW (-0.004206, $[-0.010903, 0.002288]$), and B1$-$B0 (0.001567, $[-0.005353, 0.008668]$) all include zero. The point estimates suggest that user weighting partly offsets the cost of federated optimization, but neither change is resolved overall, and B1's proximity to B0 does not show that federation has no effect, because weighting, optimizer, and training organization change together. Replacing a user's embedding with another user's drops a saved B1 seed-42 model from approximately 0.0890 to 0.0202, confirming that the model uses user-specific information.

6.2 Nonprivate low rank

The five-seed B1 mean differs from the three-seed entry above because the seed set differs. All four adjusted intervals favor B1 (table 4), so low rank incurs a utility cost before DP is introduced. The rank trend is not monotonic: rank 16 has the largest B3 mean, while rank 32 has larger seed variation and documented stability complications (appendix C). Optimization, selected rates, and stopping behavior contribute alongside capacity.

6.3 Private full matrix and low rank

B2 falls from 0.057889 in its matched no-DP control to 0.011461 at target one (table 5). Its no-DP value is already below converged B1, so the B1–B2 gap cannot be assigned entirely to noise. B4's no-DP controls also start below B2's, so a smaller loss from a weaker control does not by itself establish better recommendation quality. The sixteen adjusted B4−B2 comparisons show a conditional crossover (figure 1): all four ranks lose at target eight; no adjusted difference is detected at target four; ranks 8, 16, and 32 win at target two; and rank 8 wins at target one. Several further ordinary 95% intervals look favorable but do not survive the adjustment. The validation-selected rank at target one is 16, whose adjusted interval includes zero. At target two, B4 rank 32 reaches 0.036504 against B2's 0.025428, yet both remain below the raw-popularity reference. This motivates the private popularity comparator in E1.

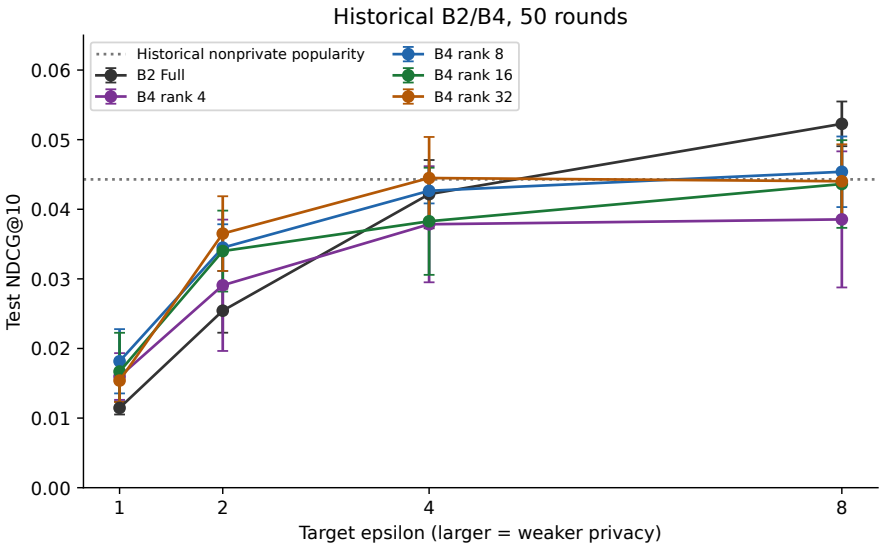


Figure 1: Five-seed test NDCG@10 with seed-SD whiskers for B2 and all B4 ranks at 50 rounds. Larger epsilon means weaker privacy. The dotted line is nonprivate raw popularity. Historical tuning budgets differ.

6.4 Communication and the original hypothesis

The rank-8 two-factor payload is 111,744 bytes per client-round against 861,184 for Full, an approximately 7.7-fold reduction: at 50 rounds, B2 averages 4.026 GB and B4 rank 8 about 0.522 GB. Converged training differs: B1 averages 93.911 GB to its best checkpoint and B3 rank 32 averages 180.329 GB, because it needs many more rounds. Ranks 4, 8, and 16 use less total traffic than B1 (15.704, 32.263, and 77.388 GB), but with lower utility. No experiment optimizes utility under an identical end-to-end communication cap.

The baselines therefore partially support the initial hypothesis: low rank can offer a better private point on the utility/payload trade-off, especially when the full model is heavily degraded. They reject its strongest form, that fewer shared noisy parameters reliably improve utility. E1 examines the proposed noise explanation directly.

Table 6: E1 test utility with ranks selected on validation at each privacy level.

ϵ	Full	Two	FixedB	DP popularity
8	0.052271	0.051158 (r32)	0.048978 (r32)	0.049548
4	0.042169	0.045807 (r32)	0.038959 (r32)	0.049428
2	0.025428	0.040841 (r8)	0.025286 (r16)	0.049507
1	0.011461	0.025136 (r4)	0.011958 (r16)	0.048389

Five-seed NDCG@10 means. The primary E1 family compares matched ranks, not these separately selected ranks. Source: test_means.csv and rank_selection.csv.

7 E1: effective noise and fixed-factor training

7.1 Why the effective matrix matters

Condition on the signal-updated factors and user matrix, and let Z_A, Z_B have i.i.d. Gaussian entries with SD τ. The immediate effective-matrix shock is

$$D = (A + Z_A)(B + Z_B) - AB = AZ_B + Z_AB + Z_AZ_B, \tag{7}$$

with expected energies (derived in appendix A.1)

$$\mathbb{E}\,\|D\|_F^2 = \tau^2(M\,\|B\|_F^2 + d\,\|A\|_F^2) + Mdr\tau^4, \tag{8}$$

$$\mathbb{E}\left\|PD^T\right\|_F^2 = \tau^2(M\left\|PB^T\right\|_F^2 + \|A\|_F^2\,\|P\|_F^2) + Mr\tau^4\,\|P\|_F^2\,. \tag{9}$$

Coordinate dimension is thus only one determinant of prediction disturbance. Modest item noise can still give a large score shock if user vectors grow, while multiplying a user's scores by a positive constant leaves their ranking unchanged. FixedB removes the terms involving a noisy B: with $BB^T = I_r$, $D = Z_AB$ and $\|D\|_F = \|Z_A\|_F$. It also changes clipping geometry and aggregation, so better conditional moments do not guarantee better trained utility. Pairwise margin variance is given in appendix A.2.

7.2 Controlled utility comparison

None of the sixteen primary comparisons favors FixedB; eight favor Two and eight include zero. At target two, FixedB is worse at all four ranks, with a rank-8 difference of -0.018662 (adjusted interval $[-0.027695, -0.010368]$). The validation-selected view (table 6 and figure 2) agrees: Two reaches 0.040841 at target two and 0.025136 at target one, against FixedB's 0.025286 and 0.011958. FixedB can perform well in no-noise controls, so the private result does not show that the implementation lacks all ranking capacity. Removing one noise source, with these transferred clipping and search choices, does not overcome the remaining learning limitations.

7.3 Private popularity changes the reference point

Bounded no-noise popularity obtains 0.049885. Its private mean at target one is 0.048389 ± 0.001125, retaining about 97% of that reference. All E1 collaborative point means at targets at most four lie below the corresponding private popularity mean, as a descriptive statement rather than an adjusted test. Popularity spends its budget on one bounded release and attempts no personalized representation learning. Its strength suggests that the useful private signal is not exhausted by the iterative methods' current optimization, though it does not show that personalization is impossible at the same target.

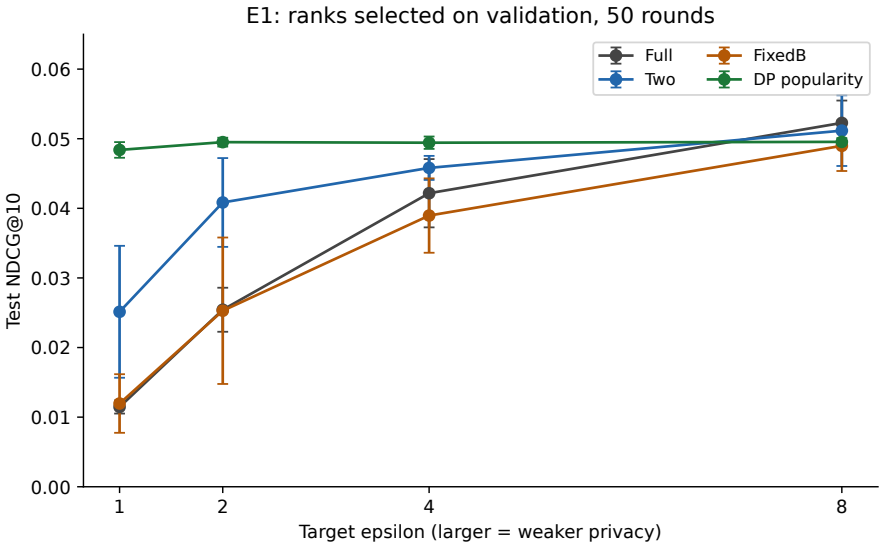


Figure 2: Full, Two, FixedB, and private popularity test NDCG@10 with five-seed SD. Ranks for Two and FixedB are selected separately on validation at each level; the primary tests compare matched ranks. Collaboration uses 50 rounds and popularity one separately calibrated release.

7.4 Measured geometry and clipping

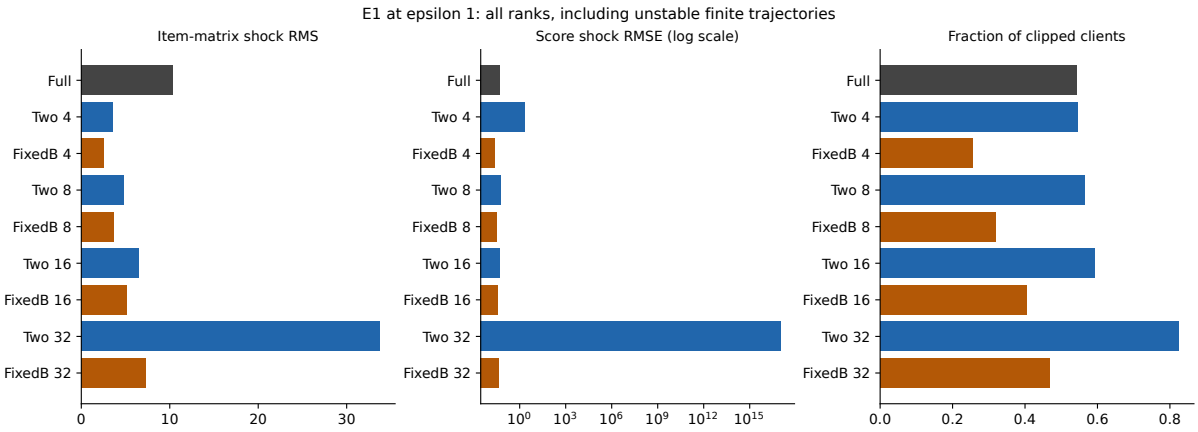


Figure 3: E1 diagnostics at target epsilon one. Left: mean over seeds of each run's RMS Frobenius norm of the immediate effective item shock. Center: mean score-shock RMSE on a logarithmic axis, retaining the large finite Two-r32 outlier. Right: selected-client clipping fraction. These are diagnostics, not causal measures of NDCG loss.

At target one (figure 3), the Full effective-noise norm is approximately 10.369; FixedB ranks 4/8/16/32 give 2.592, 3.667, 5.186, and 7.333; Two gives approximately 3.602, 4.817, 6.542, and 33.788. Lower noise does not guarantee higher utility, and selected server rates also differ. At the same target, the expected bilinear term accounts for only about 0.2–0.3% of effective-matrix noise energy in selected Two settings, which does not support calling it the dominant cause of utility failure. Clipping affects methods differently under the same threshold: approximately 54.1% of Full updates are clipped, against 25.6%–46.9% for FixedB and 54.6%–82.5% for Two, so equal C does not mean equal retained signal.

Some Two trajectories have very large but finite user norms. Two-r32's mean score-shock RMSE at target one is approximately 9.846×10^{16}, dominated by extreme state scale, so it is

not a typical-user effect. These values are reported rather than excluded (appendix C).

7.5 Activity groups, personalization, and payload

At target two, Two-r8 obtains test NDCG@10 of 0.048722, 0.027027, and 0.046489 in the low-, medium-, and high-activity groups, against private popularity's 0.055746, 0.036872, and 0.055662, so the gap is not confined to one group. Local-vector checks separate storing a user vector from benefiting from it. In the Full no-DP control, validation NDCG is 0.053136 with the user's own vector, 0.042725 with permuted vectors, and 0.047421 with the mean vector; at target one these become 0.008883, 0.011147, and 0.030485. Local persistence alone therefore does not guarantee a personalization benefit under noise. The mean-vector calculation uses private local states retrospectively and is not a deployable method; it motivates E2's explicit interventions.

Because A has $1682r$ entries and B only $64r$, fixing B saves just $64/(1682+64) = 3.67\%$ of shared coordinates relative to Two at the same rank: at rank 8, 107,648 bytes versus 111,744. Most savings relative to Full come from low rank.

8 E2/E2b: local-state memory and noise coupling

8.1 Design

During training, users adapt to the shared state, so a shared perturbation can change local vectors. E2 asks whether this pathway has measurable prediction consequences after the shared state is restored; it is a mechanism probe, not a private denoising algorithm. Starting from 30 E1 round-50 checkpoints (Full, FixedB-r8, and Two-r8; targets one and two; five seeds), each receives two seeded Gaussian pulses at amplitudes $\alpha \in \{0.25, 1\}$ times its per-round coordinate noise SD, giving 120 probes. A pulse is a fresh perturbation of a saved state, not the replay or removal of an earlier training draw. One ordinary exposure round lets selected local vectors respond. The unpulsed reference, retained-pulse branch, shared-reset branch (reference shared state, pulsed local vectors), and local-reset branch (reference local vectors, perturbed shared parameters) then continue for ten rounds on common future client, negative, and noise streams.

For reference (P, Q) and branch $(P + \Delta P, Q + \Delta Q)$,

$$\Delta S = P\Delta Q^T + \Delta P Q^T + \Delta P \Delta Q^T. \tag{10}$$

At a shared reset $\Delta Q = 0$, so the remaining difference is exactly $\Delta P Q^T$. Squared norms of the three terms do not add to the total without cross inner products, so they are not additive causal percentages. Measurements are score RMS, effective-matrix distance, top-10 disagreement on training-unseen candidates, and margin sign changes on 256 random item pairs, none using validation or test labels.

8.2 Initial findings

At full amplitude and lag ten after shared reset, Full has mean top-10 disagreement of 0.5212% at target one and 0.3206% at target two; FixedB-r8 has 0.3747% and 0.2038%. Restoring the shared state therefore does not restore every local prediction, although these modest effects do not establish a relevance loss. Raw-coordinate Two-r8 is far larger, at 17.8546% and 9.3960%. Its target-one score ratio (a probe's delayed score RMS divided by its own initial pulse RMS)

has mean approximately 911.2 but median only 1.065, indicating strong sensitivity to exceptional seeds and raising a deeper issue: the meaning of "the same noise" changes when factor coordinates are rebalanced (appendix A.5 gives a minimal example).

8.3 Orthogonal alignment control

Equivalent factors can be rotated as $(A, B) \mapsto (AR, R^T B)$ for orthogonal R without changing Q. E2b aligns a branch to the reference before each continuation round, solving the orthogonal Procrustes problem through

$$A_{\text{branch}}^T A_{\text{ref}} + B_{\text{branch}} B_{\text{ref}}^T = U\Sigma V^T, \qquad R = UV^T. \tag{11}$$

The branch then uses the same future draws in this aligned basis. Orthogonal transformations preserve isotropic Gaussian marginal laws but alter the joint coupling between branches, so a smaller paired distance does not imply a better training algorithm. Alignment needs a counterfactual reference and is an oracle control.

Across 40 declared Two-r8 probes, maximum relative product error from alignment is 3.9133×10^{-8}. At full amplitude, lag-ten disagreement falls to 0.4204% and 0.1083% at targets one and two (figure 4). The large raw distances are thus highly coupling-sensitive, and the original interpretation is not invariant to a product-preserving representation choice. Alignment is not thereby identified as the uniquely correct causal coupling. E2b is an adaptive follow-up, and six E2 and two E2b synthetic cases check reset identities, stream matching, score closure, and product preservation.

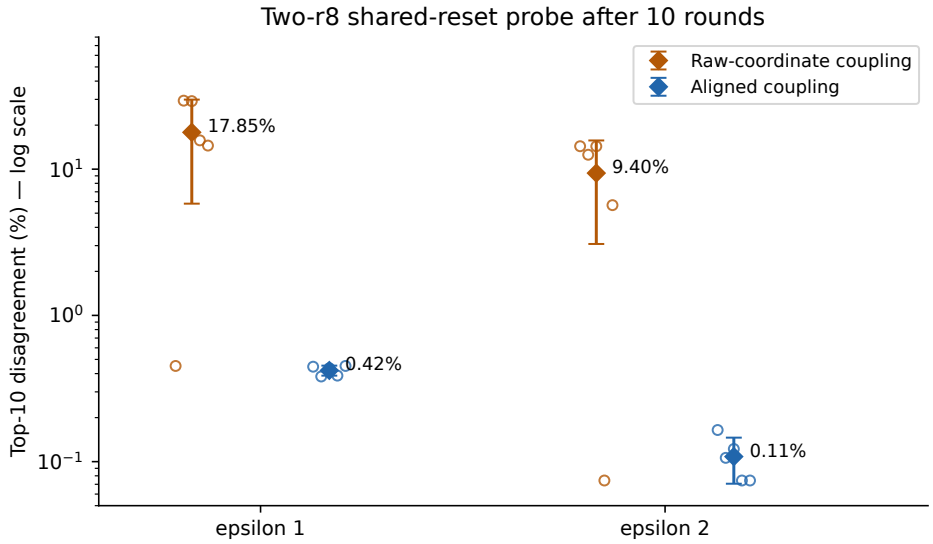


Figure 4: Two-r8 shared-reset disagreement at lag ten under raw and aligned coupling. Diamonds show means, whiskers seed SD, and circles all five seed values, on a logarithmic axis of percent top-10 set disagreement.

9 E3: larger-data replication and balancing controls

E3 trains 30 ML-1M models with the selected E1 settings and no retuning: Full, FixedB-r8, and Two-r8; targets one and two; five seeds; 50 rounds. The same noise multipliers give the same accountant bounds, while the larger qN reduces coordinate noise; dataset size also changes the

Table 7: Transferred MovieLens-1M validation NDCG@10.

Method	$\epsilon = 1$	$\epsilon = 2$
Full	0.019533 ± 0.004151	0.024624 ± 0.001777
FixedB r8	0.013661 ± 0.004574	0.016482 ± 0.004476
Two r8	0.015254 ± 0.005194	0.015440 ± 0.004567
DP popularity	0.026124 ± 0.000168	0.026161 ± 0.000104

Mean and seed SD, five seeds. Settings transfer without retuning. No ML-1M test scoring. Source: validation_summary.csv.

Table 8: E3 shared-reset top-10 disagreement at lag ten (percent of set entries).

Dataset	ϵ	SVD/raw	SVD/aligned	None/raw	None/aligned
ML-100K	1	18.6641	0.4219	0.3438	0.3438
ML-100K	2	5.2031	0.0859	0.0781	0.0781
ML-1M	1	5.2344	0.0625	0.0703	0.0703
ML-1M	2	3.3438	0.0078	0.0156	0.0156

Two-r8; 128 observer users; two draws averaged within each of five seeds. None disables balancing only in the continuation. Source: endpoint_summary.csv.

task, so NDCG differences across datasets are not a population-size effect. The main diagnostic crosses two datasets, two privacy labels, five seeds, two pulse draws, two noise couplings, and two continuation-balancing choices for Two-r8 (160 probes); 40 ML-1M Full/FixedB reference probes bring the total to 200. Pulse amplitude is one, and observers are 128 preselected users, while all users remain eligible for local updates. These streams and observer sets differ from E2, so their means need not reproduce E2's values exactly. The no-balancing condition starts from the same balanced checkpoint and disables balancing only during continuation. Unlike changing a joint coupling, this changes the continuation algorithm itself, and it is not a model independently trained without balancing.

Validation utility. All collaborative point means lie below private popularity (table 7). The fixed settings and short horizon may undertrain the larger problem, so tuned collaborative models could behave differently; this stage has no paired test and no ML-1M test score.

Main diagnostic. With SVD balancing, raw coupling produces much larger shared-reset disagreement than aligned coupling on both datasets and privacy labels (table 8 and figure 5). Without continuation balancing, raw and aligned churn are identical at saved precision for every seed, though small score RMS differences remain through float32 arithmetic. This supports a dependence on balancing-induced basis changes: the large initial Two-r8 signal is not evidence for a general failure of persistent personalization. It also does not mean balancing should be removed from an optimizer, since independently trained balanced and unbalanced methods were not compared.

The aligned effect is small in ranking terms. ML-1M Full and FixedB at target one have shared-reset churn of 0.1328% and 0.0859%, while aligned Two retains a score RMS about 3.6% of its initial pulse. In the observer sample, validation-NDCG changes under SVD/aligned Two are approximately 0.000040 and 0.000226 on ML-100K and 0.000018 and 0 on ML-1M at targets one and two; these small-sample values do not establish equivalence. All 30 runs and 200 probes

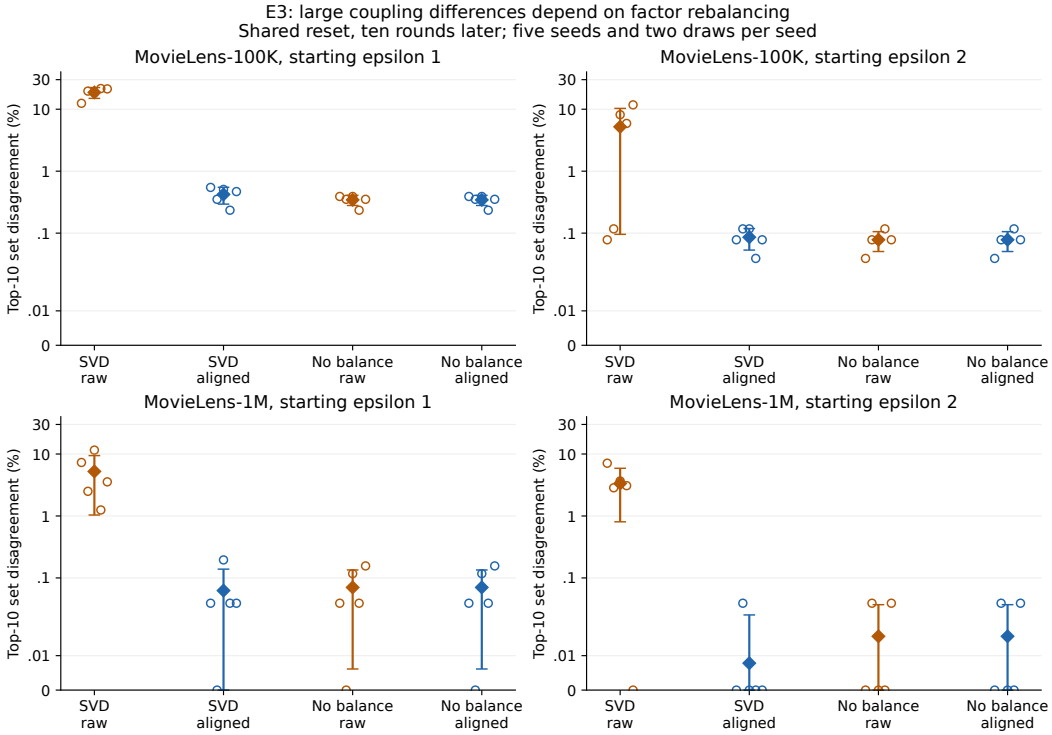


Figure 5: Two-r8 shared-reset disagreement after ten rounds on two datasets. Open circles show seed values, diamonds means, and whiskers seed SD after averaging two draws. The symmetric-log axis is linear below 0.02 percentage points so exact zeros stay visible.

complete without failures. The broad instability interpretation is rejected, and a narrower conclusion survives: representation-dependent coupling can dominate a paired noise-attribution diagnostic.

10 E4: a conditional ranking-tail screen

The bilinear term's small average energy does not guarantee accurate Gaussian tail probabilities, because ranking changes can depend on small margins. For a fixed user p, item pair i, j, and signal factors, let $v = A_i - A_j$, $u = Bp$, $s_x = \sqrt{2}\tau$, and $s_y = \tau \|p\|$. The noisy margin is

$$L = (v + s_x X)^T (u + s_y Y), \qquad X, Y \overset{\text{ind}}{\sim} \mathcal{N}(0, I_r). \tag{12}$$

With $a = v/s_x$, $b = u/s_y$, $k = s_x^2 s_y^2$, and independent noncentral chi-square U, V with r degrees of freedom and noncentralities $\|a+b\|^2/2$ and $\|a-b\|^2/2$, known distributional results [25] give (appendix A.6)

$$L \overset{d}{=} \frac{\sqrt{k}}{2}(U - V). \tag{13}$$

For a positive clean margin $m = v^T u$, numerical integration evaluates $\Pr(U \le V)$, compared with the variance-matched $\Phi(-m/\sqrt{V_{\text{lin}} + rk})$, where $V_{\text{lin}} = s_x^2 \|u\|^2 + s_y^2 \|v\|^2$, and the linearized $\Phi(-m/\sqrt{V_{\text{lin}}})$.

Screen. A declared rank-one example with $a = b = 4$ has bilinear variance share $1/33 = 3.03\%$, yet its flip probability is approximately 0.0000633405 versus 0.0026743853 for the matched Gaussian, a factor of about 42.2, so a real-state check is worthwhile. Six synthetic cases with

200,000 direct draws each check the moments and numerical integration. The screen uses 40 E1 ML-100K Two checkpoints and ten E3 ML-1M Two-r8 checkpoints, with sixteen preselected users per dataset and four training-unseen score-rank pairs, $(10, 11), (1, 11), (10, 100), (1, 100)$, giving 3,200 records. The first two pair types are primary (1,600 records). The declared gate requires matched-Gaussian absolute error above 0.01 on at least 5% of primary records in both datasets.

Outcome. The maximum matched-Gaussian error over all records is 7.66859×10^{-6}, and the maximum linearized error is 7.26162×10^{-5} (figure 6). No primary record exceeds the threshold, so the gate fails. Group-mean pairwise bilinear variance shares range from approximately 0.0011% to 0.0750%, far below the constructed example; these concern one user's item-pair margin and are not the whole-matrix energy fractions of E1. There are no numerical-failure records, and the largest quadrature error estimate is 2.64884×10^{-9}, a numerical diagnostic rather than a certified bound for every state. No amplitudes, checkpoints, users, or pair types were added after inspection.

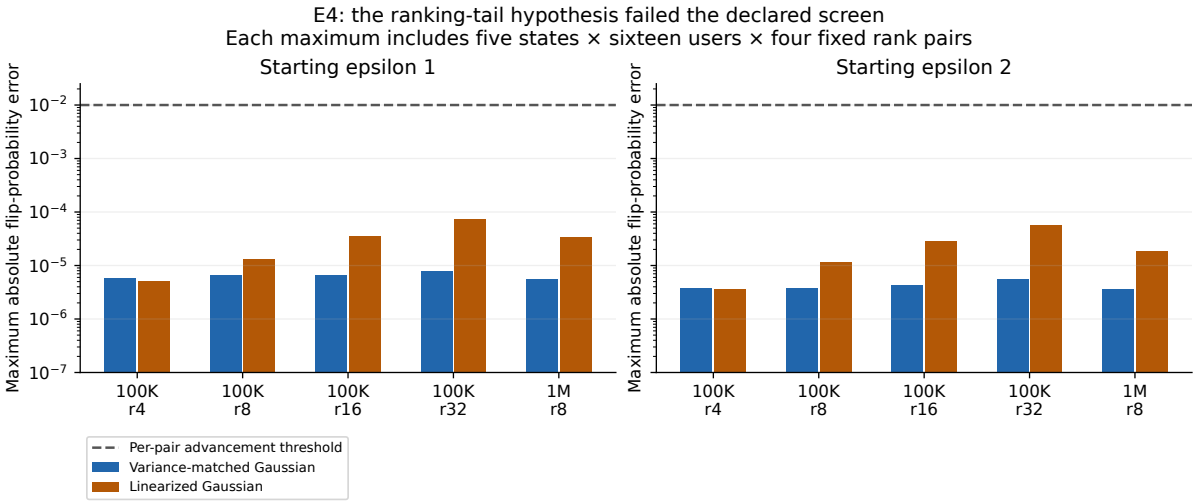


Figure 6: Maximum absolute flip-probability error for variance-matched and linearized Gaussian approximations against numerical evaluation of the conditional law. Each bar covers 320 records; the logarithmic axis marks the 0.01 per-pair gate. This is a one-step, label-free stability diagnostic, not a measure of cumulative accuracy.

The result rejects practically consequential non-Gaussian error for the sampled one-step round-50 states. It does not prove Gaussian behavior for earlier iterates, all ranks, or multi-round training, nor does it refute noisy-score stability results under explicit distributional assumptions [24].

11 Discussion and conclusions

11.1 Answers to the research questions

Table 9: Verdict and evidence for each research question.

RQ	Verdict	Evidence
RQ1 representation	Conditional gain	Four of sixteen adjusted comparisons favor low rank, all at strong privacy; every nonprivate low-rank variant is below converged B1 (section 6).
RQ2 fixed factor	Not supported	No positive adjusted comparison; eight of sixteen favor Two; private popularity exceeds all collaborative point means at target $\epsilon \leq 4$ (section 7).
RQ3 persistent state	Modest effect	Shared-reset lag-ten disagreement of 0.5212% for Full and 0.3747% for FixedB-r8 at target one; no large general relevance loss is established (section 8).
RQ4 trajectory validity	Strongly coupling-dependent	At target one, Two-r8 disagreement of 17.8546% raw versus 0.4204% aligned; raw and aligned churn are identical without continuation balancing, on both datasets (sections 8 and 9).
RQ5 ranking tails	Rejected in tested states	Maximum matched-Gaussian error of 7.66859×10^{-6} against a 0.01 gate; a larger gap appears in a constructed regime (section 10).

11.2 What the combined evidence explains

An intuitive story about private recommendation fails at several points. Reducing shared dimension lowers a coordinate-noise norm and payload, but also changes the learning geometry. Fixing a factor removes a bilinear term, yet that term's measured energy can be small while the method still loses utility. Large score disturbances can come from user scale, and large paired trajectory distances can arise from coordinate coupling rather than a comparably large ranking effect. The relevant measurements therefore form a sequence: the bounded client contribution, the effective shared signal and perturbation, the local user state, score margins, candidate rankings, and relevance. These complement the accountant, which constrains information release but does not predict usefulness, and none replaces a strong simple comparator such as private popularity.

11.3 Threats to validity

Selection and exposure. ML-100K has repeated exposure across stages. E2b is an adaptive follow-up, and E3/E4 were informed by earlier results. ML-1M validation is exposed; its test remains available.

Optimization and tuning. B4 has more tuning than B2. E1 equalizes a finite candidate count, not attainable optima, and some selected rates lie at the grid boundary. ML-1M transfer uses no retuning.

Statistics. Bootstrap intervals condition on retained runs, and five seeds say little about rare instabilities. E2/E3 probes are correlated within checkpoints and summarized descriptively,

and the small E3 observer subset limits resolution of validation differences. Group and personalization analyses are exploratory.

Privacy and systems scope. The simulator assumes noised-aggregate visibility without implementing secure aggregation. Public inputs, nonprivate selection, internal checkpoints, and diagnostics fall outside the per-run guarantee. Modeled communication excludes protocol overhead, and no deployed system is evaluated.

Generalization and numerics. Both benchmarks are MovieLens variants with unaudited overlap, so E3 is not evidence of cross-domain generalization. Per-user chronological splitting is not a global future-data experiment. Positive-threshold conversion, per-user holdout, and missing-not-at-random exposure constrain interpretation, and offline rank utility is not user satisfaction in a live service. Effective-noise formulas condition on a single state rather than describing many rounds of training, and E4 samples a finite set of round-50 margins. Reproducibility covers recorded seeds and environments, not bitwise identity across systems (appendix B.1).

11.4 Future work

1. **A published private comparator.** Implement a private ALS or federated-MF baseline with matched privacy unit, adjacency, trust model, candidate filter, and communication definition. Perturbing sufficient statistics may change the signal-to-noise regime.
2. **Public features with a private residual.** Test metadata-only local personalization, then a small private collaborative residual, against the public-only control and private popularity under an explicit total communication cap.
3. **Coupling-aware diagnostics.** Test whether they change a consequential conclusion in other settings, with the decision criterion frozen first. Any practical method must avoid counterfactual access to private reference states and receive fresh privacy accounting.

11.5 Conclusion

This project began with low-dimensional sharing as a route to better private federated recommendation and developed into a systematic study of how parameter noise becomes prediction disturbance. Its evidence comprises a reproducible benchmark, controlled fixed-factor and private-popularity comparisons, conditional noise analysis, pulse/reset experiments, a larger-data replication, and a tested tail explanation. The results support conditional low-rank gains and identify a specific limitation of factor-coordinate trajectory measurements, while rejecting the broader claims that fixed factors, smaller coordinate norms, or bilinear tails alone resolve the observed utility problem. They also identify the measurements and comparators that the next stage of this work should use.

References

[1] Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars Schmidt-Thieme. BPR: Bayesian personalized ranking from implicit feedback. In *Proceedings of the Twenty-Fifth Conference on Uncertainty in Artificial Intelligence*, 2009. URL https://arxiv.org/abs/1205.2618. Author preprint deposited in 2012; original conference paper is from 2009.

[2] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Ag"uera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pages 1273–1282, 2017. URL https://proceedings.mlr.press/v54/mcmahan17a.html.

[3] Martin Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. In *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2016. URL https://arxiv.org/abs/1607.00133.

[4] H. Brendan McMahan, Daniel Ramage, Kunal Talwar, and Li Zhang. Learning differentially private recurrent language models, 2017. URL https://arxiv.org/abs/1710.06963.

[5] Sivakanth Gopi, Yin Tat Lee, and Lukas Wutschitz. Numerical composition of differential privacy, 2021. URL https://arxiv.org/abs/2106.02848.

[6] Ilya Mironov. Rényi differential privacy, 2017. URL https://arxiv.org/abs/1702.07476.

[7] Ilya Mironov, Kunal Talwar, and Li Zhang. Rényi differential privacy of the sampled gaussian mechanism, 2019. URL https://arxiv.org/abs/1908.10530.

[8] Borja Balle and Yu-Xiang Wang. Improving the gaussian mechanism for differential privacy: Analytical calibration and optimal denoising, 2018. URL https://arxiv.org/abs/1805.06530.

[9] Xinwei Zhang, Xiangyi Chen, Mingyi Hong, Steven Wu, and Jinfeng Yi. Understanding clipping for federated learning: Convergence and client-level differential privacy. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 26048–26067, 2022. URL https://proceedings.mlr.press/v162/zhang22b.html.

[10] Steve Chien, Prateek Jain, Walid Krichene, Steffen Rendle, Shuang Song, Abhradeep Thakurta, and Li Zhang. Private alternating least squares: Practical private matrix completion with tighter rates. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 1877–1887, 2021. URL https://proceedings.mlr.press/v139/chien21a.html.

[11] Zitao Li, Bolin Ding, Ce Zhang, Ninghui Li, and Jingren Zhou. Federated matrix factorization with privacy guarantee. *Proceedings of the VLDB Endowment*, 15(4):900–913, 2022. doi: 10.14778/3503585.3503598. URL https://www.vldb.org/pvldb/vol15/p900-li.pdf.

[12] Ruixuan Liu, Yanlin Wang, Yang Cao, Lingjuan Lyu, Weike Pan, Yun Chen, and Hong Chen. PrivateRec: Differentially private model training and online serving for federated news recommendation. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023. doi: 10.1145/3580305.3599889. URL https://arxiv.org/abs/2204.08146.

[13] Alberto Bietti, Chen-Yu Wei, Miroslav Dudik, John Langford, and Steven Wu. Personalization improves privacy-accuracy tradeoffs in federated learning. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 1945–1962, 2022. URL https://proceedings.mlr.press/v162/bietti22a.html.

[14] Mihaela Curmei, Walid Krichene, Li Zhang, and Mukund Sundararajan. Private matrix factorization with public item features, 2023. URL https://arxiv.org/abs/2309.11516.

[15] Youbang Sun, Zitao Li, Yaliang Li, and Bolin Ding. Improving LoRA in privacy-preserving federated learning. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2403.12313.

[16] Ming Wen, Jiaqi Zhu, Yuedong Xu, Yipeng Zhou, and Dingding Han. Differentially private federated low rank adaptation beyond fixed-matrix, 2025. URL https://arxiv.org/abs/2507.09990.

[17] Jiawei Duan, Haibo Hu, Qingqing Ye, and Xinyue Sun. Technical report: Full version of analyzing and optimizing perturbation of DP-SGD geometrically, 2025. URL https://arxiv.org/abs/2504

.05618.

[18] Shihao Wang and Xueru Zhang. PRISM: Gauge-invariant tangent-space differentially private LoRA, 2026. URL https://arxiv.org/abs/2606.00944.

[19] Chuiyang Meng, Ming Tang, and Vincent W. S. Wong. FLoRG: Federated fine-tuning with low-rank gram matrices and procrustes alignment. In *International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2602.17095.

[20] Haoran Zhang, Dongjun Kim, Seohyeon Cha, and Haris Vikalo. FedRot-LoRA: Mitigating rotational misalignment in federated LoRA, 2026. URL https://arxiv.org/abs/2602.23638.

[21] Lele Zheng, Ruijie Hu, Tao Zhang, Ke Cheng, and Yulong Shen. FedGSA: Geometry-consistent subspace aggregation for differentially private federated LoRA, 2026. URL https://arxiv.org/abs/2608.03267.

[22] Xin Gu, Yingtai Xiao, Guanlin He, Jiamu Bai, Daniel Kifer, and Kiwan Maeng. Correlating cross-iteration noise for DP-SGD using model curvature, 2025. URL https://arxiv.org/abs/2510.05416.

[23] Xinwei Zhang, Zhiqi Bu, Borja Balle, Mingyi Hong, Meisam Razaviyayn, and Vahab Mirrokni. DiSK: Differentially private optimizer with simplified kalman filter for noise reduction. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/c7793beb7f32b559df55d48370f2b8ae-Paper-Conference.pdf.

[24] Sam Urmian, Qinyi Liu, and Mohammad Khalil. Decoupled learning and selection in slate recommendation for privacy and stability under noisy scores, 2026. URL https://arxiv.org/abs/2609.29453. Technical comparison uses the September 2026 author preprint.

[25] Robert E. Gaunt. On the product of correlated normal random variables and the noncentral chi-square difference distribution, 2024. URL https://arxiv.org/abs/2408.04101. Version 2 revised September 2025.

[26] F. Maxwell Harper and Joseph A. Konstan. The movielens datasets: History and context. *ACM Transactions on Interactive Intelligent Systems*, 2015. URL https://files.grouplens.org/papers/harper-tiis2015.pdf.

A Mathematical derivations

A.1 Conditional effective-matrix and score moments

Fix A, B, P and let the entries of $Z_A \in \mathbb{R}^{M\times r}$ and $Z_B \in \mathbb{R}^{r\times d}$ be independent with mean zero and variance τ^2. Write $D_1 = AZ_B$, $D_2 = Z_AB$, and $D_3 = Z_AZ_B$. Using $\mathbb{E}[Z_BZ_B^T] = d\tau^2 I_r$ gives $\mathbb{E}\,\|D_1\|_F^2 = d\tau^2\,\|A\|_F^2$, and likewise $\mathbb{E}\,\|D_2\|_F^2 = M\tau^2\,\|B\|_F^2$. Each entry of Z_AZ_B sums r products of independent centered variables, so $\mathbb{E}\,\|D_3\|_F^2 = Mdr\tau^4$. Cross expectations vanish by conditioning on one centered noise matrix: conditional on Z_B, $\langle AZ_B, Z_AZ_B\rangle_F$ is linear in Z_A. Expanding the squared norm proves equation (8). For scores, $\mathbb{E}[Z_B^TA^TAZ_B] = \tau^2\,\|A\|_F^2\,I_d$ gives $\mathbb{E}\left\|PD_1^T\right\|_F^2 = \tau^2\,\|A\|_F^2\,\|P\|_F^2$; similarly $\mathbb{E}\left\|PD_2^T\right\|_F^2 = M\tau^2\left\|PB^T\right\|_F^2$ and $\mathbb{E}\left\|PD_3^T\right\|_F^2 = Mr\tau^4\,\|P\|_F^2$, proving equation (9).

A.2 Pairwise margins

For distinct items i, j and fixed user p, the noisy score difference has a centered shock with variance

$$V_{ij} = 2\tau^2\,\|Bp\|^2 + \tau^2\,\|A_i - A_j\|^2\,\|p\|^2 + 2r\tau^4\,\|p\|^2\,. \tag{14}$$

The factor two arises because two independent rows of Z_A are subtracted. This shock is not generally Gaussian. For FixedB it is $(Z_{A,i} - Z_{A,j})Bp$, exactly Gaussian with variance $2\tau^2\,\|Bp\|^2$, so a clean margin m flips sign with probability $\Phi(-|m|/(\sqrt{2}\tau\,\|Bp\|))$ when the denominator is positive.

A.3 Scale changes and balancing

The map $(A, B) \mapsto (cA, B/c)$ preserves Q for $c > 0$, while the linear expected noise energy becomes $\tau^2(M\,\|B\|_F^2/c^2 + dc^2\,\|A\|_F^2)$, minimized at $c^4 = M\,\|B\|_F^2/(d\,\|A\|_F^2)$. Factor scale therefore affects measured noise even when the represented matrix is unchanged, and an equal-norm convention need not minimize it. Balancing after a noisy update preserves the already-noisy AB, so it cannot remove the realized shock. These conditional optima are not a training prescription: changing factor scale also changes client gradients and clipping, and a rule chosen from private P would need its own privacy treatment.

A.4 FixedB score capacity

If $BB^T = I_r$, define $v_u = Bp_u$. Scores are A_iv_u, with score-matrix rank at most r. Any $v_u \in \mathbb{R}^r$ has the preimage $p_u = B^Tv_u$, and other preimages add a null-space vector of B that leaves scores unchanged. Orthogonality gives $\min_{p:Bp=v}\|p\|^2 = \|v\|^2$ and $\|A_iB\|^2 = \|A_i\|^2$, so projecting the local gradient through B yields ordinary rank-r BPR dynamics for the score-relevant vector.

A.5 A product-preserving coupling null example

The pairs (A, B) and $(-A, -B)$ represent the same matrix. Applying identical raw Gaussian draws to both gives noisy products differing by

$$(A + Z_A)(B + Z_B) - (-A + Z_A)(-B + Z_B) = 2(AZ_B + Z_AB), \tag{15}$$

with expected squared distance $4\tau^2(d\,\|A\|_F^2 + M\,\|B\|_F^2)$, although both noisy products have the same marginal distribution. Coupling the second draw as $(-Z_A, -Z_B)$ makes the products

identical. A 10,000-draw synthetic check gives energy 6.341142 against theoretical 6.345313 (Monte Carlo SE 0.012443), and the aligned distance is exactly zero.

A.6 Exact conditional margin law

Let $\tilde{X} = X + a$ and $\tilde{Y} = Y + b$ in equation (12). Then $(\tilde{X} \pm \tilde{Y})/\sqrt{2}$ are independent normal vectors with identity covariance and means $(a \pm b)/\sqrt{2}$, and $\tilde{X}^T\tilde{Y} = (\left\|\tilde{X} + \tilde{Y}\right\|^2 - \left\|\tilde{X} - \tilde{Y}\right\|^2)/4$ proves equation (13). The cumulants are $\mathbb{E}L = m$, $\mathrm{Var}(L) = V_{\mathrm{lin}} + rk$, and $\kappa_3(L) = 6km$. Positive rescaling of p multiplies margin and shock alike, leaving the sign-flip probability unchanged.

A.7 Analytic Gaussian calibration for popularity

Under add/remove adjacency, equation (4) changes the unnoised sum by norm at most $\sqrt{20}$. With noise SD equal to sensitivity times σ_{pop}, the analytic Gaussian condition [8] is

$$\delta = \Phi\left(\frac{1}{2\sigma_{\mathrm{pop}}} - \epsilon\sigma_{\mathrm{pop}}\right) - e^{\epsilon}\Phi\left(-\frac{1}{2\sigma_{\mathrm{pop}}} - \epsilon\sigma_{\mathrm{pop}}\right). \tag{16}$$

Solving numerically yields multipliers of approximately 0.600229, 1.081162, 1.993812, and 3.730632 for targets 8, 4, 2, and 1. This is one release with $q = 1$, not 50 compositions of a subsampled update.

B Algorithm and settings

B.1 Client update and shared aggregation

Each local epoch is a full-batch SGD step over the client's training positives, with one fresh uniform negative per positive.

1. Sample each eligible user independently with public probability q.
2. For each selected user, copy the shared state and retrieve the local vector. Repeat $E = 2$ times: draw negatives from the training-only mask, compute the within-user mean BPR loss and regularizer, and update both the local vector and the shared copy.
3. Persist the local vector and form the difference between the shared copy and the round's starting state. For a private run, jointly clip that whole difference to norm C.
4. Sum the contributions. In private runs, add independent Gaussian noise with SD σC even if no user was selected, divide by fixed qN, and multiply by η_s. Nonprivate B1 instead averages over realized participants and skips an empty round.
5. Update the shared state and, where specified, rebalance the two factors without changing their product.

The pinned environment records NumPy 2.2.6, pandas 2.3.3, SciPy 1.15.3, PyTorch 2.14.0+cpu, Opacus 1.6.0, and matplotlib 3.10.9; Opacus is used for accountants only, not a PrivacyEngine wrapping these models. Model updates use float32, and diagnostic norms and closures use float64. A fresh-environment B0 reproduction matched reported metric precision but was not byte-identical, so reproducibility here covers recorded seeds, environments, and artifacts rather than bitwise identity across systems.

B.2 Final settings

Table 10: Final retained configurations and stopping rules.

Model	Settings
B0 / B0-UW	$d = 64$, initialization SD 0.01; Adam 10^{-3}; L2 10^{-2}; batch 1,024; maximum 1,000 epochs; patience 100. Uniform interactions / uniform users then positives.
B1	$q = 0.1$, $E = 2$, local rate 5, L2 10^{-5}, server rate 1; validation every ten rounds; patience 100 validations; maximum 8,000 rounds.
B2	B1 local settings; $T = 50$, $C = 1$, fixed qN, final-round checkpoint, PRV-calibrated noise and $\delta = 10^{-5}$.
B3	Ranks 4/8/16/32; local rates 0.625/1.25/1.25/0.625; effective-row L2; QR/core-SVD balancing; B1 stopping with the disclosed r32/seed123 ceiling exception.
B4	$T = 50$, $C = 1$, same private calibration as B2; r4 local/server 2.5/1; r8/r16/r32 local/server 5/0.5; joint factor clipping and full-coordinate noise.
E1 Full	Local/server 5/1, six levels including no-clipping/no-noise and clipping-only controls; same final iterative calibration.
E1 Two	Local 5; server 0.5 for r4/r8/r16 and 1 for r32; initial effective matrix matched to FixedB; both factors trained and balanced.
E1 FixedB	Local/server 5/1; orthonormal public basis seed 314159; private A only; 64-dimensional local vectors; no semantic metadata.
E1 popularity	Binary distinct-training-item contributions; norm bound $\sqrt{20}$; full population; one analytic-Gaussian release.
E3 transfer	E1 Full, FixedB-r8, and Two-r8 settings without retuning; two privacy targets, five seeds, 50 rounds, ML-1M normalization.

C Disclosures affecting interpretation

The following protocol revisions and incidents affect how the results should be read.

Table 11: Revisions and incidents that change interpretation.

Stage	What happened	Consequence
B2	An earlier protocol of 1,000 rounds, $C = 1.5$, and server rate 2 was replaced after validation studies by 50 rounds, $C = 1$, and server rate 1.	Three settings changed together, so the improvement cannot be attributed to horizon alone.
B3	Initial rates caused divergence. Rank 32, seed 123 used a predeclared 16,000-round diagnostic ceiling, stopping at 9,350 with best checkpoint 8,350.	Final rates and this exception are reported; all other final runs stop below the original ceiling.
B4	Inherited long-horizon rates undertrained at 50 rounds; expanded searches and a bounded fallback selected finite settings.	B4 has a larger tuning budget than B2, limiting a dimension-only comparison.
E1	Eleven search jobs failed. Float32 local-norm summaries overflowed for a finite trajectory, and float64 analysis corrected the summary.	Failures remain counted. Training, checkpoints, and test ranks are unchanged, and extreme finite results stay visible.

Table 11 (continued)

Stage	What happened	Consequence
E2b	The alignment control was declared after E2's large two-factor distances were seen.	E2b is an adaptive follow-up, and it sharply narrows the original interpretation.

D Provenance and reproduction

Evidence. Table fragments are generated from retained CSV results, with the SHA-256 hash of every input recorded; no number comes from re-evaluating saved models, and no training or test scoring was run to prepare this report. E1 records a completed 203-test suite with 26 warnings.

Data fingerprints (SHA-256).

ML-100K raw ratings	`06416e597f82b7342361e41163890c81036900f418ad91315590814211dca490`
ML-100K split	`6faed6fc3d47b5b6fa638adfeea83cd7409d50c39fa01f85c10379d0daef6989`
ML-1M split	`e68ef765cc661f5db1864f9f8b2843bee70c7ad261c116421d2c7102ec71bd54`

The ML-1M archive MD5 is `c4d9eecfca2ab87c1945afe126590906`. Raw datasets are subject to the GroupLens distribution terms.

Experiment code. The code submission that accompanies this report holds the experiment code, configurations, tests, and final result tables, and its README gives the setup steps and run order. Model checkpoints, raw run logs, and the datasets are not included, so rerunning a later stage first requires the stages it depends on. `scripts/make_figures.py` and `scripts/make_followup_figures.py` regenerate the figures from the result tables.

Companion supplement. The companion supplement holds the full retained tables, contrasts, and the detailed experimental record.